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ARAG: An Agent-Based Hybrid Semantic-Lexical Retrieval-Augmented Generation Pipeline for Arabic Texts

Mushari Alothman[a], Abdullah Altammami[a,*], Muneera Alhoshan[a], Amal Almazrua[a], Raghad Al-Rasheed[a], Rawan Almatham[a], Afrah Altamimi[a], Omar Elnashar[a], Mohamed Amin[a], Bayan Almuqhim[a], Abdulrahman Alosaimy[a], Abdullah Alfaifi[a]

[a]*king Salman Global Academy For Arabic language, Riyadh 12251, Saudi Arabia*

Abstract

Arabic Retrieval-Augmented Generation (RAG) systems face significant challenges, especially beyond hundreds of millions of words. Persistent issues include low retrieval and generation accuracy for queries involving person names, dates, religious discourse, and Arabic poetry. Additionally, high redundancy from near-duplicate content reduces accuracy and slows retrieval.

This paper introduces an Arabic-centric RAG pipeline designed to address these limitations. Evaluated on a set of 575 questions covering diverse Arabic query types, the proposed system demonstrates strong performance. The best-performing retriever, multilingual-e5-large, achieved an Acc@1 of 58.7% and an MRR of 0.715. Meanwhile, the top large language model (LLM), Gemini-2-Flash, reached an answer relevance of 0.83, a context utilization rate of 0.84, and a faithfulness score of 0.94.

Keywords: Generative AI; Retrieval-Augmented Generation (RAG); Large Language Model (LLM); Agent Retrieval-Augmented Generation (ARAG); Semantic Retrieval; Lexical Retrieval; Hybrid Retrieval; Named Entity Recognition; Metadata Filtering; BM25; Cross-Encoder Reranking; Agentic Query Routing

1. Introduction

Although Retrieval-Augmented Generation (RAG) has proven effective in English, Arabic remains underexplored. Existing systems are often limited in scale and lack openly reproducible pipelines. To bridge this gap, we introduce **ARAG** an end-to-end Arabic RAG pipeline and evaluation suite built on a large-scale Arabic text corpus. ARAG begins by applying language-specific normalization and semantic de-duplication, then segments the corpus into 512-token chunks with a 10% token overlap. Each chunk is indexed semantically in Qdrant (vector similarity search) and lexically via BM25, while a standalone NER index captures named entities. At query time, an agentic router dynamically dispatches each query to the optimal retrieval path way chunk-based, entity-based, or metadata-based

* Corresponding author .Tel.: +966-532-660-389.
E-mail address: aaltamam@ksaa.gov.sa

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then a hybrid ranker fuses Qdrant (semantic) and BM25 (lexical) scores, a cross-encoder reranker refines the top candidates. We release ARAG alongside a 575-question benchmark covering entities, dates, religious discourse, and poetry, establishing the first fully reproducible, large-scale Arabic RAG baseline.To guide our study, we formulate four research questions :

1. **Semantic-Lexical Fusion.** How does the fusion weight α in our hybrid score (Eq. 3) influence retrieval metrics (*Acc@1*, *R@20*, *MRR*) on a large--scale Arabic corpus?
2. **Reranking Benefit.** To what extent can a pair-wise cross-encoder reranker improve top-1 accuracy and mean reciprocal rank over the raw hybrid candidate list?
3. **Agentic Routing.** Does dynamically routing queries to specialised indices (chunk, NER, metadata) reduce latency or raise accuracy compared with a single-path retriever?
4. **Generation Fidelity.** Which Arabic-capable LLM delivers the most faithful and contextually relevant answers when driven by our best retriever?

Our contributions are threefold. (i) We provide a cleaned, de-duplicated, and chunk-segmented resource derived from a **large Arabic corpus** (ii) We design a hybrid semantic-lexical retrieval architecture orchestrated by an agentic controller that rewrites queries and dynamically routes them to the most relevant index whether named-entity, metadata-specific (e.g., book titles, author), or the generic chunk store, thereby maximizing recall without sacrificing latency.

The remainder of this paper is organised as follows. Section 2 surveys related work in Arabic information retrieval, Arabic-capable LLMs, and existing RAG evaluations. Section 3 introduces the *Dataset*, describing the cleaned corpus and the 575-question evaluation set. Section 4 details our *Methodology*, including data preparation, hybrid semantic-lexical retrieval, and agentic query routing. Section 5 presents the experimental setup and results for both retrieval and generation. Section 6 provides the *Conclusion*.

2. Related Work

Retrieval–Augmented Generation (RAG) combines information retrieval with large language models (LLMs) that generate answers grounded in retrieved evidence [1]. Prior work falls into two main directions: (i) general methodological advances and (ii) research specialized for Arabic.

2.1. General Advances in RAG

Agentic RAG. Classical RAG follows a single retrieve–generate cycle, whereas recent *agentic* pipelines introduce autonomous reasoning and iterative retrieval planning. [2] survey these reflection and tool-use mechanisms, illustrating how adaptive agents extend traditional RAG systems.

Semantic Retrieval. Bi–encoder models such as Dense Passage Retrieval (DPR; 3) encode queries and passages into dense vectors optimized for similarity search. Later models such as Contriever [4] achieve strong zero-shot retrieval through unsupervised contrastive training, demonstrating that scale and generalization can offset supervision.

Lexical Retrieval. Lexical approaches remain competitive for exact-match precision. BM25 [5] remains the standard baseline, while neural sparse encoders such as SPLADE [6] and COIL [7] learn context-aware token expansions that bridge dense–sparse performance gaps.

Hybrid Retrieval. Combining dense and sparse scores provides complementary gains in both recall and precision. Score interpolation between semantic and lexical signals improves performance across benchmarks [8], motivating the hybrid fusion adopted in our system.

2.2. Arabic Retrieval–Augmented Generation

Lexical and Neural Pipelines. Early Arabic information retrieval (IR) relied on stemming and diacritic normalization over BM25 [9]. The advent of contextualized encoders such as AraBERT [10] and AraELECTRA [11] significantly improved semantic matching for Arabic texts. More recent work introduced Arabic RAG benchmarks and joint retriever–generator optimization [12, 13], establishing reproducible baselines for large-scale Arabic grounded generation.

2.3. Summary and Research Gaps

Semantic, lexical, hybrid, and agentic retrieval each advance the RAG paradigm, yet Arabic research still lacks (i) large-scale corpora suitable for open research, (ii) systematic multi-metric evaluation, and (iii) studies on dynamic agentic routing. Our ARAG corpus and pipeline address these gaps and provide a replicable blueprint for future Arabic RAG studies.

3. Dataset

We work with the 322-million-word Arabic corpus and three disjoint question sets derived from the same source: (i) 575 factoid and definitional questions used for chunk-level retrieval, (ii) 110 entity-centric questions for the NER-guided retrieval pathway, and (iii) a held-out set of 100 questions for generation evaluation. The questions were first automatically generated by a large language model (ChatGPT o3) and then manually vetted by two native-speaker annotators, who checked for semantic coherence and answerability from the associated chunk, editing or discarding questions as needed. Every retained question is tagged with a subject label (e.g., علوم إسلامية, فكر إسلامي, سياسة).

3.1. Subject Distribution

Figure 1 illustrates the distribution of questions across subject categories. The dataset is heavily skewed toward *religious and cultural*—دينية وثقافية topics, with *Islamic Studies* (دراسات إسلامية) (32%) representing nearly one-third of all questions. Other prominent categories include *Hadith Sciences* (علوم الحديث) (12%), *Islamic Thought* (فكر إسلامي) (11%), *Sciences* (علوم) (9%), *History* (تاريخ) (8%), and *Journalism* (صحافة) (7%), which together account for most of the corpus. The remaining categories—such as *Novels* (روايات), *Politics* (سياسة), *Technology* (تكنولوجيا), and *Sunnah Sciences* (علوم السنة)—appear less frequently, while specialized fields like *Literary Criticism* (نقد أدبي), *Jurisprudence* (فقه), *Information Technology* (تكنولوجيا المعلومات), and *Business Administration* (إدارة أعمال) constitute only a minor fraction of the dataset.

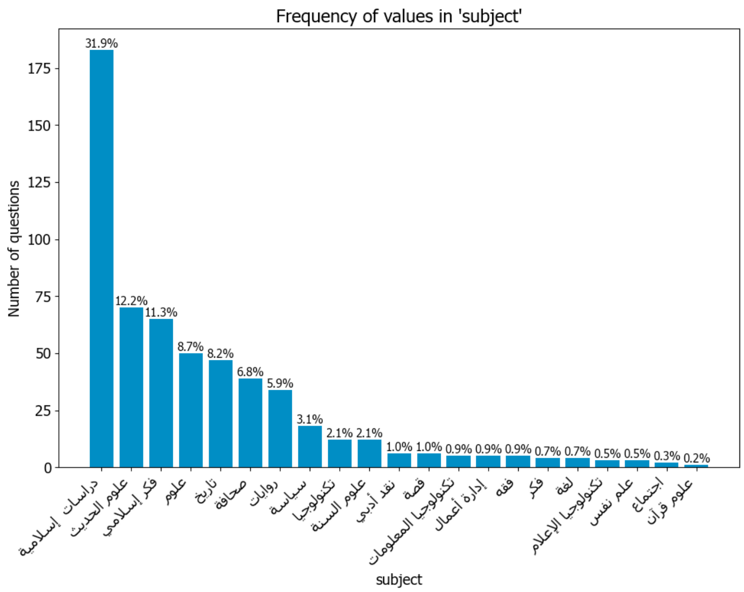

Figure 1: Distribution of the 575-question development set across subject categories.

3.2. Example Questions

1. في أي مجال يحقق الذكاء الاصطناعي نجاحًا مذهلًا؟
   In which field is artificial intelligence achieving remarkable success?
2. ما الابتكار الذي مكّن السومريين من تسجيل الضرائب؟
   What innovation enabled the Sumerians to record taxes?

4. Methodology

4.1. Overview

The ARAG pipeline comprises three main components: **data preparation**, **retrieval**, and **generation**, all orchestrated by an intelligent agentic query router. Figure 2 illustrates the data preparation stage, highlighting steps such as normalization, chunking, embedding generation, extraction of named-entity recognition (NER) and indexing. Figure 3 details the retrieval workflow, showcasing dynamic routing, hybrid semantic-lexical search, cross-encoder reranking, and iterative refinement of queries to ensure accurate and contextually relevant Arabic text generation. The following subsections elaborate each stage in detail.

4.2. Data Preparation

The data-preparation pipeline starts with two inputs the raw text corpus and its accompanying metadata files and proceeds through a single, cohesive sequence of transformations. First, every document undergoes Arabic-specific normalisation that strips diacritics, standardises punctuation, and resolves common orthographic variants. *We also unify certain letter forms—for example,* آ، أ، إ $\rightarrow$ ا*; the normalisations noticeably improves retrieval scoring.* The normalised text is then partitioned into 512-token windows with a 10% overlap; this stride preserves cross-sentence coherence while preventing transformers, whose context is capped at 512 word-pieces, from truncating salient information.

Because the corpus contains many near-duplicates that bloat the index and slow retrieval, we apply semantic duplicate suppression. For each new 512-token chunk, we compute its multilingual-E5-large embedding and compare it to previously indexed chunks. If the cosine similarity exceeds 0.95, indicating a near-duplicate, the new chunk is discarded to ensure the index remains efficient and precise.

Example at a 0.95 similarity threshold

Chunk 1 (retained)

القلبُ يضخُّ الدمَ إلى جميع أنحاء الجسمِ للحفاظ على حياةِ الإنسان.
The heart pumps blood throughout the entire body to sustain human life.

Chunk 2 (discarded)

يضخُّ القلبُ الدمَ عبر الجسم بأكمله ليبقي الإنسانَ على قيد الحياة.
The heart pumps blood across the whole body to keep a person alive.

Because the two chunks express the same fact with only minor re-wording, their cosine similarity is above 0.95, so *Chunk 2* is removed at indexing time.

In parallel, we extract named entities with `marefa-nlp/marefa-ner`[1] and retain all publisher-supplied metadata (Person, Location, Organization, Nationality, Job, Product, Event, Time, Art-Work). These discrete tokens are stored

[1] https://huggingface.co/marefa-nlp/marefa-ner

in dedicated Elasticsearch indices, enabling fast keyword-based BM25 lookup over entity and metadata fields. The resulting artefacts are stored in two back-end databases. For each chunk, its dense embedding is written to a Qdrant *vector database*, while the same chunk's lexical representation— together with all NER and metadata fields—is indexed in *Elasticsearch* for high-precision BM25 lookup. This dual-store layout mirrors Figure 2 and forms the retrieval backbone for all downstream RAG components.

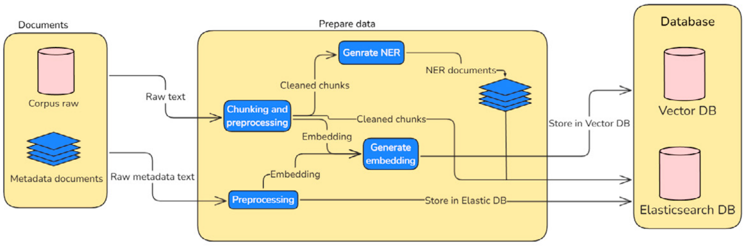

Figure 2: Data preparation: from raw corpus and metadata ingestion through chunking, NER, embedding, and dual indexing in databases.

4.3. Agent Router

Upon receiving a user query, the **agent router** implemented with the `Gemini-2-Flash` model and a rule-based controller classifies the query into one of three categories:

- **Metadata-related**:
  Queries targeting document-level metadata (e.g. author, publication date, topic tags).
- **NER-related**:
  Queries focusing on named entities (persons, organisations, locations) extracted during preprocessing.
- **Chunk-based**:
  Queries requiring full-text retrieval over the 512-token chunks.

Each category is backed by its own vector store (Qdrant) and lexical index (Elasticsearch). All retrievals employ the hybrid score of Eq. 3. Depending on the classification result, the agent routes the query through the appropriate pipeline, while the rule-based fallback logic guarantees a maximum of two retrieval passes to ensure coverage and robustness.

4.4. Search

Figure 3 illustrates the end-to-end search workflow within the ARAG pipeline. At its core, an intelligent agent analyzes the user's query and routes it into one of three specialized retrieval pathways:

1. **NER Retrieval:** Targets named-entity queries. The query is preprocessed, passed through a BM25 search, and reranked to retrieve the top relevant entity chunks.
2. **Chunk-Based Retrieval:** Handles general semantic and lexical queries. The query is preprocessed and embedded, then parallel BM25 (lexical) and Qdrant (semantic) searches are executed. Results are fused and reranked via a cross-encoder to yield the top chunk IDs.
3. **Metadata Retrieval:** Deals with document-metadata queries. It follows the same dual BM25 and Qdrant vector search and fusion process, but focused on metadata fields.

Top-ranked chunks from these pathways are then fed to dedicated LLMs:

- **Chunk-Based LLM:** Generates answers using the retrieved chunks; if the response is insufficient, an iterative query-refinement loop is triggered.
- **Metadata LLM:** Specializes in metadata queries, with a fallback to the chunk-based LLM when necessary.

This adaptive routing ensures that ARAG dynamically selects the best retrieval strategy for each Arabic query, maximizing both relevance and accuracy.

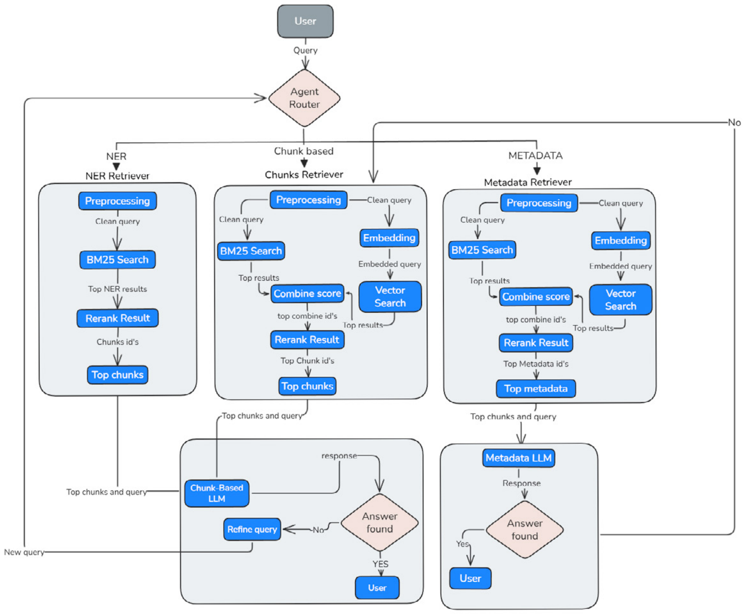

Figure 3: Query workflow: agentic routing, hybrid retrieval (NER, chunk, metadata), cross-encoder reranking, and iterative query refinement.

4.5. Scoring Functions

BM25. For a query q containing terms t, the sparse score is

$$\mathrm{BM25}(q,c) = \sum_{t\in q\cap c} \mathrm{IDF}(t)\, \frac{f_{t,c}\,(k_1+1)}{f_{t,c} + k_1\left(1 - b + b\,\frac{|c|}{\overline{|D|}}\right)} \tag{1}$$

with term frequency $f_{t,c}$, document length $|c|$, collection average length $\overline{|D|}$, and $\mathrm{IDF}(t) = \log(1 + \frac{N-n_t+0.5}{n_t+0.5})$. We keep the canonical hyper-parameters $k_1 = 1.2,\ b = 0.75$

Cosine similarity. Sentence embeddings $e_q, e_c \in \mathbb{R}^d$ are ℓ_2-normalised, so the cosine similarity reduces to a dot product:

$$\cos(e_q, e_c) = \frac{\langle e_q, e_c\rangle}{\|e_q\|_2\,\|e_c\|_2} = \langle \hat{e}_q, \hat{e}_c\rangle. \tag{2}$$

Vector normalisation. Before indexing, we ℓ_2-normalise each embedding so that $\|\hat{e}\|_2 = 1$. In this form, the dot product $\hat{e}_q^\top \hat{e}_c$ computed by the vector database is numerically equivalent to cosine similarity (see Eq. 2), eliminating the need to divide by vector lengths at query time. Performing normalisation at indexing time allows the system to compute similarity through a single multiply-and-add operation per vector pair, yielding substantial efficiency gains compared to computing cosine similarity on the fly.

4.6. Hybrid Search with Weighted Fusion

Retrieving Arabic text passages demands both semantic alignment and precise lexical matching. We therefore score every query-chunk pair (q, c) with

$$S_{\mathrm{hybrid}}(q,c) = \alpha\, \cos(e_q, e_c) + (1-\alpha)\, \mathrm{BM25}(q,c) \tag{3}$$

where the fusion weight $\alpha \in [0, 1]$ controls the balance between semantic (dense) and lexical (sparse) signals, with higher values emphasizing semantic similarity.

4.7. Metadata Filter

Objective. Enable domain-focused retrieval by restricting search to chunks that match a user-specified metadata key value pair (e.g., `book_name=` التفسير بالبيان المتصل في القرآن الكريم, `vessel_name=` رسائل علمية)). Filtering occurs *before* hybrid scoring (Eq. 3), improving both precision and latency.

Implementation.

1. **Metadata Mapping.** During indexing, each chunk is tagged with all available metadata and stored in a bidirectional dictionary for fast lookup. In our system, the metadata filter is *user-scoped only*we do not infer or manually annotate query domains. Documents carry domain labels, which are inherited by their constituent chunks. At retrieval time, users may optionally restrict search to a specific domain or source (e.g., book). If no filter is applied, the retriever searches the full corpus. This design serves as a practical hygiene mechanism in mixed-domain corpora (e.g., preventing religious texts from being retrieved for technology-related queries), without assuming any prior knowledge of where the answer resides.
2. **Dual Retrieval & Fusion.** At query time we apply the user-supplied metadata filter to both stores, retrieve the top candidates independently from *(a)* Qdrant (semantic) and *(b)* Elasticsearch (lexical), and then combine the two result lists with the hybrid semantic–lexical scoring in Eq. 3 before final ranking.

5. Evaluation

In this section, we separately evaluate the retrieval and generation stages of ARAG. For retrieval, we measure the impact of semantic-lexical fusion weights, reranking, and named-entity indexing on top-1 accuracy (Acc@1), recall at 20 (R@20), and mean reciprocal rank (MRR). For generation, we assess how faithfully Arabic-capable LLMs produce answers grounded in retrieved contexts, using the RAGAS framework.

5.1. Retrieval

We systematically examine several key factors influencing ARAG's retrieval performance. Specifically, we first identify the optimal semantic-lexical fusion weight (α) that balances semantic and lexical retrieval signals. Next, we measure the effectiveness of cross-encoder reranking in refining retrieval accuracy. Finally, we evaluate the named-entity retrieval pathway separately, analyzing its impact on precision and latency. The following subsections detail each experimental evaluation.

5.1.1. Choosing the Fusion Weight α

To determine the optimal balance between semantic and lexical signals in our hybrid search (Eq. 3), we evaluate several fusion weights α, and Table 1 summarizes results for three semantic-to-lexical ratios. A semantic-heavy mix ($\alpha = 0.8$) clearly achieves the best retrieval performance, improving Acc@1 by approximately 12 points and R@20 by roughly 29%, confirming that a small but significant lexical component substantially improves retrieval accuracy by compensating for Arabic morphological variation and common spelling inconsistencies. Our hybrid thus improves over pure BM25 by incorporating semantic understanding through a language-model encoder that is better suited for Arabic dialectal text: BM25's exact term matching alone often fails on dialectal synonyms and morphological/orthographic variants (e.g.,«مال» vs. «فلوس»), whereas dense embeddings capture their semantic equivalence while BM25 contributes complementary lexical precision.

Table 1: Impact of semantic–lexical fusion weight α on retrieval metrics

α (semantic:lexical)	Acc@1	R@20	MRR
1.0 : 0.0	46.7%	67.6%	0.538
0.8 : 0.2	**58.7%**	**97.0%**	**0.715**
0.5 : 0.5	56.5%	88.5%	0.675
0.0 : 1.0	50.43%	78.83%	0.593

5.1.2. Effect of Cross-Encoder Reranking

We rerank the top-20 hybrid retrieval set ($\alpha = 0.8$) with a pair-wise cross-encoder and compare performance before and after reranking in Table 2. Reranking lifts Acc@1 by roughly 10–12 percentage points for the multilingual-E5 family while leaving R@20 almost unchanged (>89% for all E5 variants), showing that the hybrid retriever already offers strong coverage and that the cross-encoder adds clear value by substantially improving the final ranking precision.

Table 2: Retrieval performance before and after cross-encoder reranking (hybrid search with $\alpha = 0.8$).

Model	Dim	Acc@1		R@20		MRR	
		without rerank	rerank	without rerank	rerank	without rerank	rerank
multilingual-E5-large	1024	46.2%	**58.7%**	96.52%	**97.0%**	0.5842	**0.715**
multilingual-E5-base	768	41.2%	57.9%	90.4%	91.6%	0.535	0.693
bge-m3	1024	41.0%	57.3%	89.2%	90.2%	0.529	0.684
multilingual-E5-small	384	38.6%	56.1%	89.0%	90.0%	0.517	0.679
GATE-AraBERT	768	28.1%	48.7%	73.3%	74.9%	0.396	0.575
mpnet	768	14.6%	27.4%	39.4%	40.7%	0.204	0.320
AraBERT-v2	768	7.6%	19.4%	31.6%	32.8%	0.127	0.235

5.1.3. Named-Entity Retrieval Evaluation

For the 110-question entity-centric benchmark, we evaluate retrieval using the dedicated NER index rather than the general chunk index. Because the entity questions are typically short and highly specific, the named-entity pathway performs strongly: it attains a top-1 accuracy of **62.7%**, a recall@10 of **94.5%**, and an MRR of **0.736**. This corresponds to roughly a 16% point gain in Acc@1 over the general chunk index under the same protocol. Named entities are extracted with `marefa-nlp/marefa-ner` and retrieved solely via Elasticsearch BM25, confirming that a specialised NER pathway is particularly effective for entity-focused Arabic queries.

5.2. Generation

5.2.1. Motivation

Although Arabic retrieval quality has improved substantially (Section 5.1), the usefulness of RAG ultimately depends on whether generated answers are factual and grounded in retrieved evidence. In large-scale Arabic settings, this remains challenging: hallucination rates are higher than in English, dialectal variation and code-switching are widespread, and orthographic and morphological variability complicate both retrieval and generation. Moreover, hybrid semantic–lexical retrieval and reranking pipelines optimize similarity scores but do not guarantee that the model conditions on the retrieved passages instead of relying on its internal knowledge. To address this gap, we adopt a focused evaluation framework, instantiated via RAGAS, that jointly measures factual faithfulness with respect to retrieved spans, answer relevance, and context utilisation; we additionally validate and calibrate these automatic scores with targeted human judgments to confirm that retrieval improvements translate into reliably grounded Arabic answers.

5.2.2. Experimental Setup

We benchmark five publicly available large language models (LLMs) with competitive Arabic capability: Gemini-2-Flash, Qwen2.5-Instruct-7B, Qwen3-8B, c4ai-command-r7b-arabic-02-2025, and Gemma3-12B. For each query, we provide the top-20 evidence chunks retrieved by our best hybrid retriever (Section 5.1). The chunks are embedded in a structured prompt that (i) explicitly instructs the model to base its response solely on the provided evidence, (ii) requires the output to be returned as a JSON object of the form {"answer": ...}, and (iii) instructs the model to return "no answer" when the evidence is insufficient or not relevant to the query. Decoding is performed with temperature 0.1 and a 2000-token generation budget, balancing determinism and expressiveness.

5.2.3. Metrics

Automatic evaluation is conducted with RAGAS [14], a reference-free framework recently adopted by several Arabic RAG benchmarks [13]. We report three complementary dimensions:

- **Faithfulness**: natural language inference based entailment between each atomic claim in the answer and at least one retrieved context span.
- **Answer Relevance**: semantic congruence between the generated answer and the user query, judged by an expert LLM classifier.
- **Context Utilisation**: proportion of answer tokens that can be aligned to any retrieved evidence span, approximating grounding adequacy.

Table 3: Generation evaluation of Arabic-capable LLMs on a 100-question validation set.

Model	Faithfulness	Answer Relevance	Context Utilisation
Gemini-2-Flash	**0.94**	**0.83**	0.84
Qwen2.5-Instruct-7B	0. 86	0.70	0.81
Qwen3-8B	0.90	0.68	**0.87**
c4ai-command-r7b-arabic-02-2025	0.41	0.30	0.40
Gemma-3-12B	0.30	0.25	0.33

5.2.4. Limitations

This study is subject to four main limitations.

1. **Prompt Sensitivity**. A single prompt template was employed; systematic prompt-engineering or chain-of-thought variations remain for future work.
2. **Evaluation Bias**. RAGAS relies on judge models that may inadvertently encode Arabic-specific or genre-specific biases. Although we partially mitigated this risk through human adjudication, automatic scores will inevitably drift as judge models evolve.
3. **Format Compliance**. Several models sporadically ignored the required `JSON` schema, producing free-form prose or malformed keys.While post-processing heuristics corrected most outputs, residual parsing errors could influence both automatic and manual assessments. Developing stronger format-conditioning techniques (e.g. constrained decoding) is therefore warranted.

5.3. Discussion

Retrieval findings: Taken together, Tables 1 and 2 show that a semantic-heavy fusion ($\alpha = 0.8$) delivers the best balance between semantic coverage and exact matching, boosting *Acc@1* by roughly 12 percentage points and *R@20* by about 29 percentage points over a semantic-only baseline. On top of this, cross-encoder reranking yields substantial additional gains in top-1 precision, while recall remains virtually unchanged. This pattern indicates that the hybrid retriever already provides strong coverage and that reranking effectively sharpens the ordering of candidates rather than expanding the candidate set. The agentic router's NER pathway further achieves a class-leading 62.7 % *Acc@1* in Section 5.1.3, confirming the value of intent-aware routing and specialised indices for real-time, entity-centric Arabic queries.

Generation findings: As shown in Table 3, Gemini-2-Flash derives substantial benefit from the high-quality retrieval pipeline, surpassing the next-best model by several points in Faithfulness while also maintaining strong Context Utilisation. Its comparatively modest advantage in Answer Relevance suggests that grounding quality—rather than intrinsic generative capacity alone—is the decisive performance factor in our setup. Conversely, the poor performance of *c4ai-command-r7b-arabic-02-2025* and *Gemma-3-12B* underscores the importance of robust instruction-tuning on multi-sentence, evidence-centred justifications, a data regime still under-represented in openly available Arabic corpora. Overall, the generation results mirror the retrieval findings: models that are better able to exploit the retrieved evidence, rather than merely relying on parametric knowledge, deliver the most reliable Arabic answers.

6. Conclusion

This paper introduced ARAG, an agentic retrieval-augmented generation pipeline tailored to large-scale Arabic text. ARAG shows that integrating a hybrid semantic-lexical retrieval architecture together with an intelligent agentic query router substantially improves both retrieval precision and generation quality. Key components of our pipeline include specialized retrieval pathways such as Named-Entity Recognition (NER) retrieval, metadata-filtered search, and a sophisticated cross-encoder reranker to refine retrieval accuracy. The system's best-performing retriever achieved an R@20 of 97% and an MRR of 0.715, while the Gemini-2-Flash language model demonstrated superior performance with a faithfulness score of 0.94, answer relevance of 0.83, and context utilization rate of 0.84.

The implications of these results are significant for Arabic language processing, addressing critical challenges such as vocabulary mismatch, semantic ambiguity, and the complex linguistic characteristics of Arabic, including dialectal variation and rich morphological structure.

Despite these advancements, several limitations must be acknowledged.The framework demonstrates strong potential for multilingual adaptation, though its general applicability remains to be further validated through cross-domain and cross-language experiments.

Future research should focus on refining these components to handle linguistically challenging contexts more robustly, exploring advanced prompt engineering and chain-of-thought reasoning to further enhance fidelity. Additionally, extending the evaluation to include other high-performing language models and commercial Arabic-supportive models as they become available could provide deeper insights. Developing specialized datasets for complex linguistic domains and promoting interdisciplinary collaborations will further advance the robustness and applicability of RAG systems.

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